THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 11: **Keeping Time with the Skies** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): The shadow of a vertical stick is shortest at noon. Reason (R): At noon the Sun is highest in the sky. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. A pole of height 3 m has a shadow of 4 m and a nearby tree has a shadow of 20 m at the same time. The height of the tree is: 1

(a) 10 m **(b)** 20 m **(c)** 15 m **(d)** 26 m

3. The Moon appears the same size as the Sun in the sky although the Sun is about 400 times larger because the Sun is also: 1

(a) closer **(b)** about 400 times farther
(c) smaller **(d)** dimmer

4. Assertion (A): A leap year has 366 days. Reason (R): The Earth takes about 365.25 days to complete one revolution around the Sun. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. A student measures the shadow of a 1 m stick as 0.5 m at noon on one day and 1 m a few months later. This shows the Sun's noon elevation has: 1

(a) become zero **(b)** decreased **(c)** increased **(d)** stayed same

6. A calendar with 12 months of 29.5 days is adjusted by adding a month every: 1

(a) every year **(b)** every 10 years **(c)** every month **(d)** about 3 years

7. Assertion (A): We always see the same side of the Moon. Reason (R): The Moon does not rotate on its axis. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. A total lunar eclipse is seen at midnight from a place. At that time the Moon is: 1

(a) not present in the sky **(b)** high in the sky
(c) below the horizon **(d)** close to the Sun

9. After 33 years the lunar and solar years differ by about: 1

(a) one month **(b)** one day **(c)** ten years **(d)** one year

10. Assertion (A): The Moon is visible at night. Reason (R): The Moon produces its own light. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does a calendar need a leap-day rule with exceptions for century years? Give an example with 1900 and 2000. 2

12. State the reason why the Moon cannot be seen at the time of a new moon, although it is above the horizon in the daytime. 2

13. How did the discovery of the Earth's revolution explain the changing night sky through the year? 2

14. If a 1.5 m stick has a 2 m shadow, find the height of a building with a shadow 40 m long at the same time. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. At a place the Sun is directly overhead at noon on 21 March. A pole of height 5 m is placed there. Describe the shadow at noon and at 3 pm. 3

16. Why do countries have different time zones? How many hours of difference per 15 degrees of longitude? 3

17. Explain the length of the Earth's orbit and the speed of the Earth in orbit (use 150 million km radius, circumference 2 pi r) in km/s approx. 3

OR

A year consists of 365.2422 days. Calculate how many days the calendar will be ahead after 100 years if 365.25 is used and say how the Gregorian rule corrects it. 3

18. India is 82.5 degrees E for IST. Find the time difference between a place at 75 degrees E and IST. 3

OR

How do the shadows of a pole at the equator and at a high latitude differ at noon in June? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe a long-term investigation to record the Sun's noon shadow length and direction through the year and explain the expected results with the solstices and equinoxes. 5

OR

A traveller sees the Pole Star at 30 degrees altitude. State his latitude, say where the Pole Star would appear at the equator and at the North Pole, and what happens to its height as he travels north. 5

20. Explain, with calculations, how the Gregorian calendar corrects for the 365.2422-day tropical year, listing the three rules for leap years, and test the years 1800, 1900, 2000, 2024 and 2100. 5

OR

Explain the changes in the night sky through the year and how Orion, Ursa Major and Cassiopeia can be located, describing the shapes. 5

END OF QUESTION PAPER — Total: 40 Marks